

**Success of Ecological Restoration Projects in Austin, Texas, Based on the Urban
Greenspace Integrity Index**

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Abstract

Urbanization and its consequence of habitat degradation threaten native biodiversity. To mitigate these damages, cities worldwide have implemented ecological restoration. Studying the success of restoration projects provides valuable information on the effectiveness of environmental policies, especially for fast-growing cities, which are more susceptible to the negative impacts of urbanization. For this reason, this study analyzes Austin, Texas, one of the fastest-growing cities in the United States with 15% of its land dedicated to urban green space (Connolly and Lira, 2021). This study conducts a city-level comparison of 10 Austin ecological restoration projects to determine their relative success. To achieve this, this study calculated the Urban Greenspace Integrity Index (UGII) of 10 Austin green spaces with restoration projects. Introduced by Callaghan et al. (2019), the UGII is a “locally-relevant measure of an urban green space’s response to urbanization, derived from widely accessible citizen science data.” The results demonstrate that among the 10 selected green spaces, Commons Ford Ranch Metro Park and the Vireo Preserve had a UGII greater than 1, meeting the study’s definition of a “successful” restoration project that represents the increase of urban-sensitive species. Mayfield Park was the least successful restoration project out of the ten. For the City of Austin, this study emphasizes the importance of addressing specific restoration needs and avoiding generalizing restoration successes across different green spaces.

Keywords: Austin, Texas, Urbanization, Urban restoration, Ecological restoration, Urban Greenspace Integrity Index, Urban tolerance, Citizen science data, eBird

Introduction

The United Nations (2019) predicts that by 2050, 68% of the world's population will live in urban areas. Increased land-use intensity, population growth, and urbanization significantly contribute to habitat and biodiversity loss (Simkin et al., 2022). However, cities worldwide have strived to mitigate these damages and revitalize degraded ecosystems through ecological restoration (McCann & Schusler, 2017). The Society for Ecological Restoration (SER) defines ecological restoration as “the process of assisting the recovery of an ecosystem that has been degraded, damaged, or destroyed” (SER, 2004). In addition to improving urban biodiversity, successful restoration projects connect people with nature, provide environmental education, and improve citizens' physical and psychological well-being (Dearborn & Kark, 2010). Thus, it is imperative to assess the progress of these projects and determine the current effectiveness of restoration efforts in fast-growing cities. Such research would provide a better understanding of the resilience and sustainability of urban ecosystems in a future of environmental challenges such as climate change, population growth, and continued urban expansion.

This study compares the progress of different ecological restoration projects in Austin, Texas. By analyzing the effectiveness of the environmental efforts of a rapidly growing city, local practitioners and policymakers can gain a deeper understanding of urban ecosystems and the future implementation of restoration projects.

Literature Review

Goals for Ecological Restoration

There is extensive debate surrounding the idea of a “successful” restoration. The primary guideline for restoration success, the SER Primer (2004), provides a list of attributes encompassing species composition, ecosystem function, landscape context, and ecosystem

stability, suggesting that restoration sites should develop these attributes to be “successful” (Shackelford et al., 2013). In recent years as restoration ecology progressed, studies have challenged this guideline; for example, Shackelford et al. (2013) and Hallett et al. (2013) suggest the inclusion of human well-being as a restoration goal. Nevertheless, the attributes introduced in the Primer are still widely accepted internationally.

One attribute for successful restoration suggested in the Primer is the presence of native species (SER, 2004). This indicator is crucial in urban areas, where disparities from natural ecosystems and structural homogeneity promote biotic homogenization -- the trend where native species are driven out and replaced by non-native species (McKinney, 2006). A reason for this trend is that while native species are unaccustomed to disrupted ecosystems, exotic species possess traits that help them thrive in urban habitats; for example, the rock dove evolutionarily adjusted to rocky environments and thus preadapted to concrete buildings (McKinney, 2006). In addition to biotic homogenization, invasive urban-tolerant species can harm native species through competition, hybridization, and disease transmission (Martin-Albarracin et al., 2015). Archibald et al. (2017) support McKinney’s work and address the concepts identified by Martin-Albarracin et al. by suggesting that urban restoration should be incredibly attentive to “urban sensitive species,” defined as “species whose abundances tend to be highest in undisturbed habitats.” Callaghan et al. (2019) further confirm the works of McKinney and Archibald et al., emphasizing the importance of restoring urban green spaces so that urban-sensitive species with lower urban tolerances, or lower “urbanness,” can re-enter the community.

Measurements of Restoration Outcomes and Diversity

Similar to the characteristics of successful restoration, the methods to measure restoration outcomes vary immensely as well. To evaluate the measures used in the field, Ruiz-Jaen and Aide (2005) compared articles that assessed restoration projects based on the attributes introduced in the SER Primer (2004). Their review concludes that the three major measured attribute categories are “diversity, vegetation structure, and ecological processes” (Ruiz-Jaen & Aide, 2005). Adding on to the work of Ruiz-Jaen and Aide, Wortley et al. (2013) reviewed publication trends, finding that diversity was the most frequently measured category out of the three.

A large body of literature has discussed species diversity as a measurement of restoration outcomes. Roswell et al. (2021) discuss traditional diversity metrics such as species richness and the Shannon index in their guideline for measuring species diversity. First, species richness “measures the number of species” in an ecosystem, so the more species there are, the more diverse the area is (Roswell et al., 2021). However, not only is species richness not robust against outliers (i.e., species that are present but only make up a small percentage of the total population), richness measured at smaller samples does not effectively predict the relative richness in larger samples (Roswell et al., 2021). Contrarily, the Shannon index describes diversity by considering “the proportion of each species” (Konopiński, 2020). Thus, even if the species richness of an ecosystem is equal, the index would be able to “[distinguish] between sites dominated by a single or only a few predominant species and those where each species has comparable input to the whole biodiversity” (Konopiński, 2020). In addition to these traditional measurements, other species diversity metrics include Hill diversity and Rényi’s entropy (Hill, 1973; Rényi, 1961). Callaghan et al. (2019) acknowledge these diversity indices but point out

that they fail to consider scenarios where more harmful and urban-tolerant non-native species replace native species. Thus, even if diversity metrics do not detect differences between communities, the “urbanness” of local species is crucial to evaluate.

Regardless of the metric used, studies that monitor restoration usually compare the project sites to a “reference ecosystem,” defined by the SER Primer (2004) as a site representing the ideal ecosystem conditions the restoration project expects to emulate. Thus, reference sites are crucial in planning and evaluating the outcomes of a restoration project. In their review, Wortley et al. (2013) found that 74% of restoration studies “included some form of reference or control site for comparison.” Studies that did not have reference sites mostly “tracked the development of the restoration site over time instead” (Wortley et al., 2013). However, for urbanized areas, reference sites can be challenging to identify because instead of returning to an original, undisturbed state, restoration projects would benefit from “[building] upon the nature that already exists within cities” (Callaghan et al., 2019; Gobster, 2010). As for long-term studies that monitor restoration, “lack of funding” and “the relatively short working lifespan of the researchers” make them challenging to conduct (Lindenmayer et al., 2012). Therefore, methods of restoration evaluation that do not require reference sites would be substantially more accessible for city planners.

These studies demonstrate that the ecosystem attributes, diversity metrics, and reference sites used for restoration assessment depend greatly on project contexts. However, not all methods are appropriate for urban settings due to differences in reference conditions, urban tolerance of species, and additional factors such as funding. Thus, it is crucial to utilize methods suitable for urban areas to provide a better understanding of restoration outcomes in light of population growth and continued urbanization.

The Urban Greenspace Integrity Index

One specific way to measure the effectiveness of restoration efforts is through the Urban Greenspace Integrity Index (UGII). Introduced by Callaghan et al. (2019), the UGII measures individual species' responses to urbanization and “quantifies ‘improved diversity,’” taking into account the presence of “desired species,” defined as species “particularly vulnerable to anthropogenic disturbances.” The UGII does not require a reference site to measure restoration, making it more affordable and applicable for urban green spaces.

This metric is similar to beta diversity, defined as the ratio between regional and local species diversity (Tuomisto, 2010; Whittaker, 1960). As a standardized metric, beta diversity allows for comparisons between disparate locations, making it a helpful way to measure community biodiversity differences (Socolar et al., 2016). Instead of diversity, however, the UGII is defined as the proportion of regional and local-level urbanness (Callaghan et al., 2019). Therefore, the UGII allows for cross-comparisons of restoration success among locations with different regional species pools.

Moreover, the UGII utilizes citizen science as a data source. Citizen science data is collected by volunteers of the general public and has improved data quality and communication between data users and collectors (Sullivan et al., 2014). Huddart et al. (2016) support Sullivan et al. (2014) and describe how citizen science data is an effective solution to the “lack of robust standardized data” and can advance the field of restoration science. In addition to these benefits, citizen science data is cost-effective and “disproportionately collected near urban areas,” making it especially useful for tracking restoration progress in cities (Callaghan et al., 2019). Kosmala et al. (2016) assert that volunteers can collect data of the same quality as professional ecologists; thus, biodiversity scientists should implement citizen science more widely.

The UGII can utilize observation data from various species, but Callaghan et al. (2019) specifically examined bird species. Much of pre-existing research has assessed ecosystem conditions based on the responses and diversity of bird species. Temple and Weins (1989) explain the wide use of birds as indicators, describing how birds respond predictably to environmental changes, “especially aspects of environmental degradation that may not yet be apparent to us.” Burnett et al. (2005) agree with the work of Temple and Weins, adding that the cost-effectiveness and relative ease of studying avian species make them “a model organism for measuring the success of restoration.” Overall, the UGII applies to urban areas by considering species’ urban tolerance, using cost-effective data collection, and being comparable across different locations.

Gap

Austin is one of the fastest-growing cities in the United States and is also traditionally known as “Texas’ most green and ecological city, with 15% of land within city limits dedicated to parks and other green spaces and an abundance of regional nature reserves” (Connolly & Lira, 2021). Nevertheless, there is an apparent lack of research on ecological restoration in Austin in the fields of urban ecology and restoration ecology. Furthermore, few studies have utilized the Urban Greenspace Integrity Index to compare different restoration projects across a city. The UGII, tailored towards urban areas, would provide a valuable overview of Austin’s environmental efforts and shed light on the relationship between future urban expansion and biodiversity.

Methodology

Overview

This study hypothesized that all Austin ecological restoration projects chosen would not be successful since Austin is rapidly urbanizing (City of Austin, 2021). Increased human activity within the green spaces due to population growth would hinder restoration if not further degrade the local ecosystem. To test this hypothesis, this study calculates the Urban Greenspace Integrity Index (UGII) for ten green spaces in Austin, Texas. This project's overarching purpose is to determine the relative effectiveness of ecological restoration projects in Austin, potentially providing insight into which specific Austin projects require more investment and which types of projects are the most effective in urban environments similar to Austin. This understanding will be increasingly important as climate change and urban expansion continue.

Study Locations

Austin was chosen for this study because of its large amount of green spaces, parks, and trails, extensive sustainability measures, and long history of environmental preservation (Connolly & Lira, 2021). Ten green spaces that have implemented ecological restoration projects were chosen for this analysis (See Figure 1 and Table 1). For this study, ecological restoration will follow the definition of the SER Primer (2004), and Table 1 shows examples of the activities considered.

Figure 1

Map of Study Locations

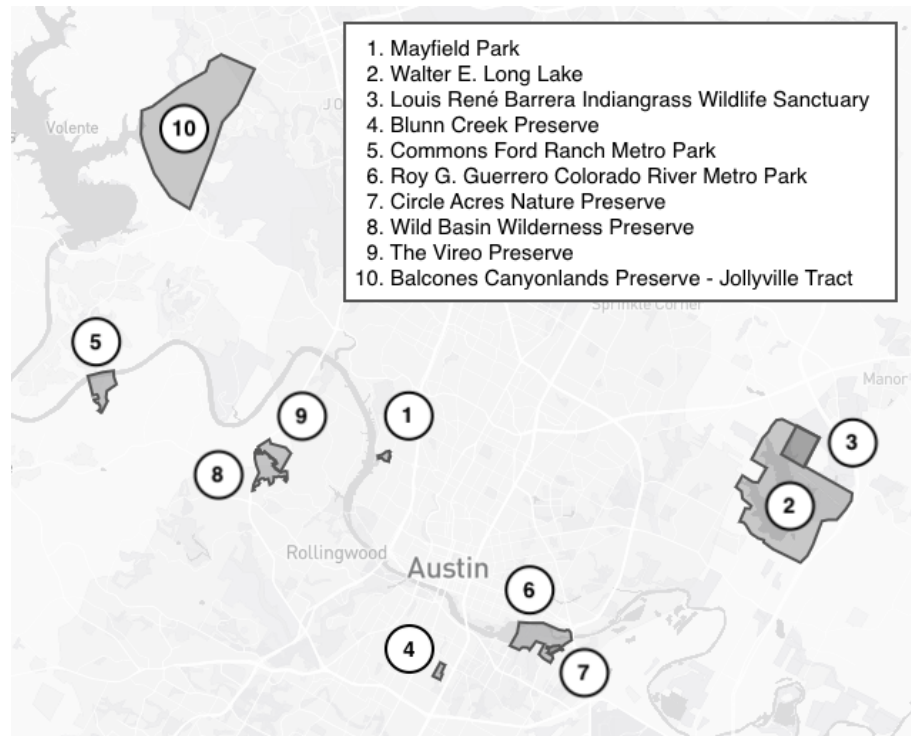


Table 1

Chosen Green Spaces and Some of Their Respective Restoration Efforts

Green Space	Restoration
Mayfield Park	Invasive species removal (Siglo Group, 2015).
Walter E. Long Lake	Invasive species removal and woody plant encroachment (Austin Parks Foundation, 2020).
Louis René Barrera Indiangrass Wildlife Sanctuary	Three large restoration projects were completed, including prescribed fire (Austin Parks Foundation, 2020). Located within Walter E. Long Park.
Blunn Creek Preserve	Invasive species removal (Austin Parks Foundation, 2020).
Commons Ford Ranch Metro Park	Invasive species removal (Austin Parks Foundation, 2020).
Roy G. Guerrero Colorado River Metro Park	Biodiversity enhancement and water quality improvements (Texan by Nature, 2021).

Circle Acres Nature Preserve	Restoring a former dump and illegal landfill into an urban wildlife sanctuary (Texan by Nature, 2021). Located within Roy G. Guerrero Park.
Wild Basin Wilderness Preserve	The Wild Basin Wilderness Preserve, The Vireo Preserve, and the Jollyville tract are all managed under the Balcones Canyonlands Preserve (BCP). The BCP is “one of the nation’s largest urban preserves,” with “more than 140 individual tracts managed by both public and private partners” (Travis County, 2019). For the Vireo Preserve specifically, projects are implemented to promote soil restoration and native plant regeneration (The Balcones Canyonlands Conservation Plan, 2022).
The Vireo Preserve	
Balcones Canyonlands Preserve -- Jollyville Tract	

To minimize the inaccuracies of manually delineating the green space boundaries, borders were downloaded from iNaturalist, another citizen science project and widely popular global biodiversity recording platform (Callaghan, Ozeroff, et al., 2020). As a result, the green spaces chosen 1) must be listed as a recognized place in iNaturalist; 2) must be listed as a hotspot in eBird to ensure a complete set of data listed under this location; 3) must be within an iNaturalist boundary that mainly corresponds with a single eBird hotspot. This mitigates potential inaccuracies of only using one set of eBird data while the boundaries encompass multiple sets.

Data Source

This study uses bird observation data from the eBird Basic Dataset (see Appendix A). eBird is a prominent citizen science project that “collects information about the distribution and abundance of birds,” hosted by the Cornell Lab of Ornithology (Sullivan et al., 2014). eBird collects data in the form of checklists submitted by volunteer birdwatchers; checklists document observation information such as time, date, and location. Complete checklists are those where all birds identified visually or audibly are recorded. eBird was chosen because there is extensive eBird data in Austin, and Callaghan et al. (2019) utilize this platform in their methodology.

All eBird observations in Texas, USA, from January 1, 2010, to the latest available data at the time of the data collection (November 30, 2022) were downloaded. This period was chosen as January 2010 marks the growth of the eBird platform, providing a more extensive, less biased sample (Callaghan et al., 2019).

The monthly time frame used for this study was springtime in the Northern Hemisphere (March 15 to June 15). The reason behind this time frame is the variation in biodiversity due to seasonal differences: only about 16% of all Texas birds are resident species that do not have any seasonal movement (Shackelford & Lockwood, 2000). However, the UGII does not apply to regions with substantial seasonal variations in species diversity (Callaghan et al., 2019). Thus, only data from the approximate Texas breeding season was chosen for this study.

Calculating UGII

Figure 2

Workflow of UGII for Each Restoration Project

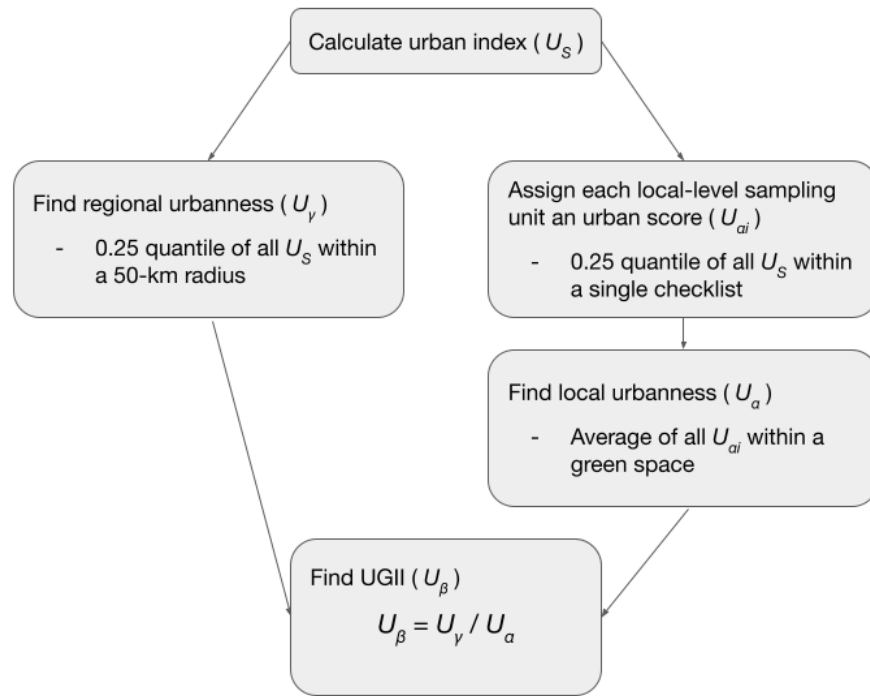


Figure 2 summarizes the process of finding the UGII for restoration projects. First, an urban score (U_s) – a number quantifying a species’ urban tolerance – is calculated for each species. The higher the U_s , the more urbanized the species is. Hence, restoration projects should aim to attract birds with lower urban scores, which reflects the green space as a safe environment for urban-sensitive native species. Second, a zone with a 50-km radius – the project’s “regional” area – would be delineated around each site. The 0.25 quantile of the distribution of U_s in this “region” would be the location’s regional urbanness (U_r). The purpose of using the 0.25 quantile is to “encompass a minimum level of ‘less urban birds’” (Callaghan et al., 2019).

Meanwhile, the 0.25 quantile of the distribution of U_s in each checklist – also taken to encompass less urban-sensitive birds – would represent the urban index of each individual eBird checklist (U_{ai}). Third, the average of all U_{ai} within the site boundaries would be the local urbanness (U_a). Finally, the UGII (U_β) would be calculated by dividing the regional urbanness by the local urbanness. A green space with a UGII close to or greater than 1 would be defined as “successful” since the proportion indicates that local species are equally or less urban-tolerant than regional species, signifying a green space with fewer anthropogenic disturbances.

Urban Index

Prior to calculations, the bird species’ urban scores (U_s) must be obtained. These indices are calculated by assigning each observation with the mean value of a continuous measurement of urbanization (i.e. night-time lights) within a 5-km buffer, then finding the median urbanization value within the distribution of observations. However, since most urban scores of Texan birds needed for this study were already calculated and published in the study “Urban tolerance of birds changes throughout the full annual cycle” conducted by Callaghan, Cornwell et al. (2021), all urban indices used for this project were obtained from that study (see Appendix A).

This set of indices from Callaghan, Cornwell et al. included monthly scores to account for the monthly variation in species urbanness, but since only the Texas breeding season was considered in this analysis, the average score from March to June was taken for each species.

Calculating regional urbanness (U_r)

Initially, the eBird data was converted into a MariaDB database utilizing the SQL scripts provided by Callaghan et al. (2019) (see Appendix A). Mariadb is an open-source relational database (mariadb.org) that can filter large datasets and determine whether or not a coordinate fits within a certain area, making it ideal for this study. Additionally, GeoJSON files were created for the sites' boundaries and 50-km buffer zones (created in dropchop.io). Then, in R Studio, eBird checklists located within the buffer zones were selected from the Mariadb database (see Appendix A).

After obtaining the regional data, only species that appeared on more than 2.5% of the checklists were included. This lowers the likelihood of including uncommon birds that are not considered part of Austin's usual regional species pool. Then, the 0.25 quantiles of each 50-km zone were calculated to find the regional urbanness (U_r).

Calculating local urbanness (U_a)

To calculate U_a , the data within the 10 green spaces' boundaries were extracted. To ensure that the data is of the highest quality and to avoid any outliers that might skew the results, this study followed the methodology used by Callaghan et al. (2019) and filtered the pool of eBird observations based on the following criteria:

- I. Include only checklists where birding was the observer's primary focus. This is validated by only including complete checklists and checklists that followed the eBird birdwatching protocols of "traveling," "stationary," or "area."

- II. Include only checklists that recorded birds between 5 and 240 minutes. This removes any outliers in observation time.
- III. Include only checklists that traveled < 5 km or < 500 Ha. Narrowing the range of bird observations prevents accidentally associating data with a faraway green space.

Theoretically, U_{α} would be the simple average of all U_{ai} – the 0.25 quantile of the urban scores in each individual checklist – within a green space. However, to account for typical sampling biases in citizen science data (e.g. the correlation between time spent birding and the number of birds observed), generalized additive models were fitted, which calculated a single “average” checklist that minimized the impacts of these biases, then predicted U_{α} based on that “average” checklist (Callaghan et al., 2019).

Calculating U_{β}

The final UGII result was calculated by $U_{\beta} = U_{\gamma} / U_{\alpha}$. See Appendix A for the complete set of R codes used for this methodology.

Results

A total of 407,565 eBird observations were used for the regional scale and 2,356 for the local scale after filtering, encompassing 129 bird species. Table 2 shows the final calculations.

As seen in Table 2, of the 10 green spaces analyzed, Mayfield Park had the lowest UGII of approximately 0.705, and Commons Ford Ranch Metro Park had the highest of about 1.117, with the range being about 0.4. All locations chosen had a UGII greater than 0.7, demonstrating that all restoration projects are relatively successful. The green spaces with the most similar UGII were the Louis René Barrera Indiangrass Wildlife Sanctuary (hereafter referred to as the Indiangrass Preserve) and Roy G. Guerrero Park, with a difference of 0.001, meaning they had the most similar levels of restoration effectiveness. Both Commons Ford Ranch and the Vireo

Preserve had a UGII greater than 1, which is an exceptional score that exceeded this study's hypothesis.

Table 2

Final Calculations Ranked From Highest UGII to Lowest

Location	Ugamma	Ualpha	Ubeta
Commons Ford Ranch Metro Park	2.095737	1.875988	1.1171377
The Vireo Preserve	2.089434	2.026124	1.0312465
Balcones Canyonlands Preserve--Jollyville Tract	2.091535	2.164674	0.9662123
Roy G. Guerrero Colorado River Metro Park	2.091535	2.277832	0.9182128
Louis René Barrera Indiangrass Wildlife Sanctuary	2.105818	2.297404	0.9166074
Walter E. Long Lake	2.093636	2.352021	0.8901435
Circle Acres Nature Preserve	2.093636	2.402769	0.8713432
Blunn Creek Preserve	2.093636	2.646458	0.7911087
Wild Basin Wilderness Preserve	2.089434	2.752305	0.7591578
Mayfield Park	2.091535	2.965823	0.7052123

Discussion

In response to the research question, the results reveal that 2 of the ten selected Austin green spaces have achieved successful ecological restoration, rejecting the hypothesis.

Although there is no discrete range for a “successful” restoration project, a UGII greater than or equal to 1 is more desirable because it reflects that the bird species in a restoration area are more or similarly urban sensitive compared to the birds in the region, which represents a green space more equivalent to the natural environment. Hence, the results demonstrate successful restoration in Commons Ford Ranch and The Vireo Preserve. Since the other green spaces with UGII scores less than 1 attract a disproportionate amount of urbanized birds, they

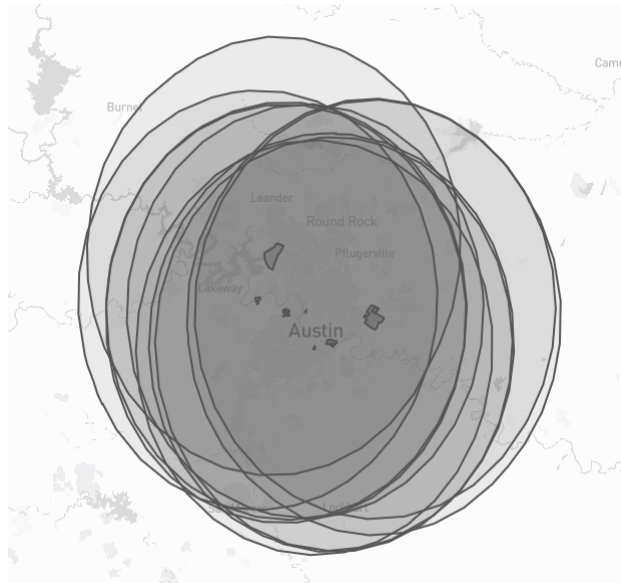
should focus on improving or maintaining existing restoration efforts to allow their UGII to increase over time and reach 1.

Regional and Local Urbanness

All locations had a regional urbanness between approximately 2.106 and 2.089 – only a 0.017 range. A small difference is expected because, as seen in Figure 3, since the green spaces are within close proximity, the 50-km zones would mostly overlap; hence, the regional species pool is fundamentally the same for all Austin projects.

Figure 3

50-km Zones Around Each Green Space Delineating Their Regional Species Pool



Due to the similarity in regional urbanness, local urbanness alone successfully predicted the UGII ranking of the study sites – the higher the local urbanness, the lower the UGII. This affirms the predictions of Callaghan et al. (2019), who expected U_α to behave similarly to U_β for green spaces in the same city. However, though local urbanness sufficed as a comparison metric for these Austin areas, it is still imperative to place local urbanness in context to account for any variation in the regional species pool.

Next, it is essential to note the difference in findings for the three green spaces managed under the BCP program. While the Vireo Preserve and Jollyville tract had the second and third highest UGII of around 1.031 and 0.966, respectively, the Wild Basin Wilderness Preserve had the second lowest UGII of around 0.759. It is also striking that even though the Vireo Preserve and Wild Basin Preserve are directly adjacent, their results differ by 0.27. Adjacent locations could have attracted the same species, but the environmental conditions of individual green spaces greatly impact local biodiversity. Hence, the two locations' contrasting UGII underscores the importance of addressing individual restoration needs of green spaces, especially in large preserves such as the BCP that may lack specialized management. Governments must be cautious of overgeneralizing successes and appropriately direct restoration funding.

Moreover, the results show a minimal difference in effectiveness for “interior” green spaces (i.e., the Indiangrass Preserve and Circle Acres) and their external counterparts. The Indiangrass Preserve is located within Walter E. Long Park, and their UGII only differed by 0.03, with the internal project being more effective. Contrarily, Circle Acres's UGII was 0.05 lower than Roy G. Guerrero Park's, meaning restoration efforts outside Circle Acres were more effective than the ones within it. With no observable pattern on the relationship between the restoration effectiveness of internal and external projects, this study concludes that they are similarly effective for Austin green spaces.

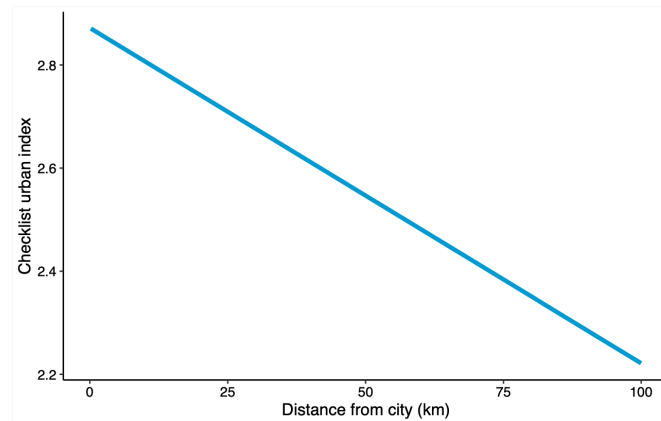
Urban Tolerance of Austin Birds

The species with the lowest urban scores analyzed were the Brown-crested Flycatcher (0.38), Northern Bobwhite (0.57), and Vesper Sparrow (0.60); conversely, the three most urban species were the Monk Parakeet (a high outlier with a score of 19.91), House Sparrow (8.48), and Black-crowned Night-Heron (7.13).

Initially, this study's data justifies the use of urban scores in Austin because, as Figure 4 demonstrates, urban indices are negatively correlated with the distance from the urban center, supporting the work of Callaghan et al. (2019).

Figure 4

Urban Index on a Checklist

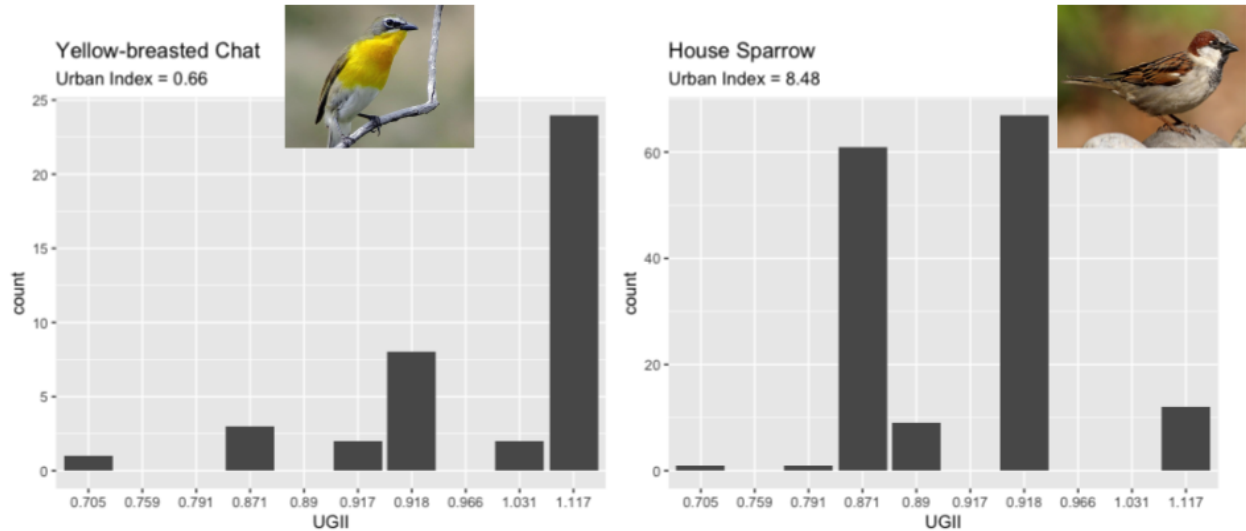


Note. The graph displays the relationship between the distance (km) from Austin's city center and the urban index on a checklist (U_{ai}), showing a linear model fit of the data.

The data also affirms that some birds do, in fact, prefer specific green spaces over others, and Commons Ford is more capable of attracting urban-sensitive species. Figure 5 displays how each species' preferences influenced the UGII score: less effective projects generally attracted fewer urban-sensitive birds (i.e., Yellow-Breasted Chat) and more invasive urbanized birds (i.e., House Sparrow). The Yellow-breasted Chat has a low urban index of 0.66, while the invasive House Sparrow has a high urban index of 8.48.

Figure 5

Two Examples of Species Distribution Across Austin Green Spaces



Note. This figure displays two examples of the number of times a bird species appeared in each green space chosen for this study.

Therefore, Austin restoration projects should monitor these discrepancies and adapt their restoration methods accordingly to benefit native species instead of invasive, urban-tolerant species.

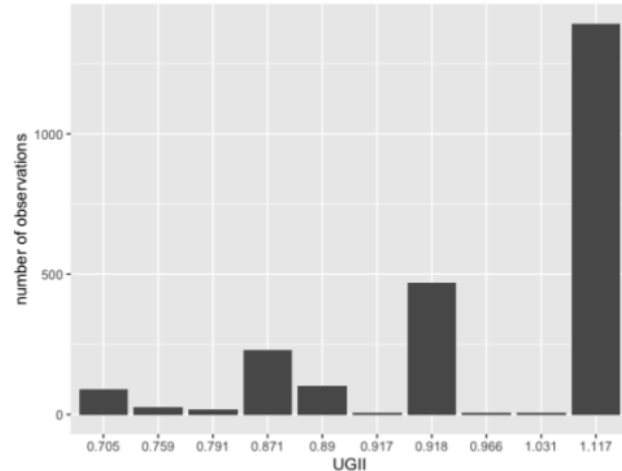
UGII and Sample Size

Among the 2356 observations used for this study, 1393 were located in Commons Ford Ranch, accounting for 59% of the total dataset. The observations that met this study's criteria in Commons Ford – a popular birding hotspot – significantly exceeded other green spaces. Thus, this study portrays the scenario where citizen science data is unequally collected for study sites.

Figure 6

Number of Observations in Each Green Space Ordered By UGII

Green Space	Count
Commons Ford Ranch Metro Park	1393
The Vireo Preserve	8
BCP--Jollyville	8
Roy G. Guerrero Colorado River Metro Park	471
Louis René Barrera Indiangrass Wildlife Sanctuary	6
Walter E. Long Lake	104
Circle Acres Nature Preserve	231
Blunn Creek Preserve	18
Wild Basin Wilderness Preserve	27
Mayfield Park	90



As seen in Table 3 and Figure 6, observation numbers and data collection efforts can vary significantly across sites, justifying the use of UGII over simple species richness. This variation would be more pronounced when comparing green spaces continentally or globally, as factors such as socioeconomic status influence the amount of citizen science data available (Callaghan, Poore, Mesaglio, et al., 2020). Thus, generalizable and comparable measures such as the UGII should be used when assessing restoration with variable data to minimize sampling error.

Conclusions and Policy Implications

This study assessed the effectiveness of Austin ecological restoration projects and found that among the selected sites, Commons Ford Ranch and the Vireo Preserve were exceptionally effective based on the UGII. This study contributes to previous literature on applying citizen science data in measuring restoration and determining species responses to urbanization. Additionally, this study adequately addresses the current gap in literature, providing a city-level comparison of multiple restoration projects in Austin by calculating species urbanness to determine the relative successfulness of each green space.

The City of Austin must recognize green spaces still highly impacted by urbanization, such as Mayfield Park, and prioritize restoring ecosystems and protecting urban-sensitive native species. As for more successful restoration projects such as those of Commons Ford and The Vireo Preserve, the state of the environment should be maintained to prevent future damage.

Limitations

One limitation of this study is the separation of urban score calculations from actual UGII calculations. Although Figure 4 demonstrates that generally, species with lower scores are further from Austin's city center, possible discrepancies still may exist between bird species' urban tolerance on the continental scale, as found by Callaghan, Cornwell et al. (2019) and on the local scale of Austin, potentially affecting the results.

Another limitation is the wide variation in study site sample sizes. Table 3 shows that Commons Ford Ranch had significantly more observations compared to other locations, which might have contributed to its result as the most effective project. As for locations such as the Indiangrass Preserve, with only six observations, the UGII may have been unrepresentative of the sites as a whole due to small samples.

Furthermore, the time frame of the final UGII spans from 2010 to 2022, meaning that it represents overall restoration progress without accounting for inter-annual differences within these years. For example, if a site's restoration substantially improved during 2022, this analysis would not have factored in this change and thus may overlook the most recent restoration progress.

Finally, citizen science platforms such as eBird have inherent biases. Data is more frequently collected on weekends, migration periods, and mornings, in areas with more accessibility and higher biodiversity, and are often of larger-bodied birds in larger groups

(Courter et al., 2012; Sullivan et al., 2014; Tiago et al., 2017; Geldmann et al., 2016; Callaghan, Poore, Hofmann, et al., 2021). As a result, this study's conclusions may be unrepresentative of the actual progress of Austin's restoration.

Future Research

This study catalyzes many areas of future research, the most significant one being the factors of green spaces or specific restoration efforts that led to a high or low UGII for Austin green spaces. Additionally, future research should consider the variety of UGII applications provided by Callaghan et al. (2019), including annual tracking, global comparison, and assessing larger, city-level restoration projects instead of local-level green spaces.

Finally, to address this study's limitations, future research can calculate species-specific urban scores using Texas data to apply urbanness measurements on a smaller scale. Future research can also analyze sites using larger sets of bird observations, incorporate data from other citizen science projects such as iNaturalist, track city green spaces annually with the UGII, or expand beyond Austin and compare other major Texas cities such as Dallas and San Antonio. All areas could address the current gap in the literature on urban restoration in Texas.

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Appendix A

Urban Greenspace Integrity Index Calculation Procedure and Resources

All code can be accessed from “Using citizen science data to define and track restoration targets in urban areas” (Callaghan et al., 2019). Additional steps taken as well as modifications to the code in the context of Austin will be explained below.

1. Convert eBird data to the SQL database with “init.sql”
 - a. Before the conversion, if a MacBook is used as in this analysis, install the Mariadb server through Homebrew.
 - b. Before the conversion, download eBird data from Texas (Jan 1 2010 to November 2022). eBird data is available at <https://ebird.org/data/download/ebd>.
 - c. Certain variables present in Texas eBird data differ from those in the Australian eBird data. The variables are: “TAXON CONCEPT ID,” “EXOTIC CODE,” “BREEDING CODE,” “BREEDING CATEGORY,” “BEHAVIOR CODE,” “STATE,” “STATE CODE,” and “COUNTY CODE.” Change these variables before importing the data to the SQL database.
2. Choose study locations, then create boundaries and buffers.
 - a. After comparing the areas in iNaturalist and eBird based on this study’s criteria, find the iNaturalist “place” at <https://www.inaturalist.org/observations/identify>.
 - b. Substitute “#” in the following URL with the iNaturalist place ID:
<https://www.inaturalist.org/places/geometry/#.kml>
 - c. Download the KML file of study location boundaries.
 - d. Convert the file to GeoJSON with “kml_to_geojson.R” (Edwards, 2013).
 - e. Create 50-km buffer zones in dropchop.io.

3. Spatially query the SQL database based on the GeoJSON files with
“spatial_query_get_lists_in_buffer.R”
4. Find the urban index based on Callaghan calculations from the study “Urban tolerance of birds changes throughout the full annual cycle” using “urban_index.R”
 - a. This is an additional code that should find the average score from breeding season
5. Determine which counties the buffer zones cross based on Texas County Boundaries from TxDOT (available at <https://gis-txdot.opendata.arcgis.com>), and download eBird data from those counties. Directly downloading county data instead of splitting the Texas data into individual data frames would save time.
6. Read in data and convert to RDS with additional code titled “read_ebird_data.R”
7. Get $U\gamma$ with “get_u_gamma_for_each Greenspace.R”
8. Clean up the data to prepare for analysis with “clean_data_script.R”
9. Get $U\alpha$ with “get_ualpha.R”
 - a. This is originally titled “analysis_script.R”; however, most analysis done by Callaghan et al. (2019) is not needed for this study, only the $U\alpha$ calculation was used.
10. Find the final result ($U\beta$) with “plot_metrics.R”